Qassim University
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جامعة القصيم
كلية الحاسب
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Social media impact on teen mental health

PROJECT REPORT

CS435

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PROJECT OVERVIEW

Objective:

This project focuses on predicting depression among teenagers using Machine Learning techniques. Because of the increasing effects of social media, stress, anxiety, academic pressure, and psychological challenges on adolescents, mental health has become an important issue that requires early detection and proper awareness.

The main objective of this project is to develop predictive machine learning models that can identify whether a teenager is likely to experience depression based on social, behavioral, and academic factors. The project also aims to analyze the relationship between these factors and teenage mental health in order to better understand the causes and indicators of depression.

In addition, this project seeks to demonstrate how Machine Learning can assist in supporting mental health analysis, improving prediction accuracy, and contributing to early intervention and prevention strategies for teenagers at risk of depression.

Expected outcomes:

The project aims to build a Machine Learning system that can predict depression among teenagers using factors such as stress, anxiety, academic performance, social interaction, and social media usage.

It also compares the performance of two models, **Logistic Regression** and **Decision Tree** Classifier, to determine which provides better prediction accuracy using evaluation methods like accuracy score, classification report, and confusion matrix.

In addition, the project is expected to:

- **Identify** factors related to teenage depression.
- **Show** the importance of data preprocessing and SMOTE balancing techniques.
- **Visualize** the Decision Tree model for better understanding.
- **Support** early detection of mental health risks and increase awareness of teenage mental health.

DATASET

Dataset Source:

The dataset used in this project was obtained from Kaggle and focuses on teenagers’ mental health and the impact of social media usage on depression levels.

Description:

Size: The dataset contains (13) columns and approximately (1,200) rows.

Variables: The dataset includes several variables such as:

- **Demographic** data (such as: age, gender)
- **Social media** usage (such as: daily social media hours,social media platform usage)
- **Academic factors** (such as: academic performance)
- **Social interaction** level
- **Mental health indicators** (such as: stress level, anxiety level, depression label)

Data Types: The dataset consists of different data types, including:

- **Numerical** data (such as: age, stress level, anxiety level)
- **Categorical** data (such as: gender, platform usage)
- **Binary** classification target (depression label: depression / no depression)

Data Preparation:

The dataset underwent several preprocessing and cleaning steps to improve data quality and ensure reliable results before applying machine learning models.

• First, **missing values** were identified using null checks, and incomplete records were removed. **Duplicate** rows were also detected and deleted to ensure that each record represents a unique observation.

• Next, data types were **standardized** for consistency. Numerical variables such as stress_level and anxiety_level were standarized for consistency and prepared for model training.

• **Categorical** data was also **cleaned and standardized**. For example, the gender column was standardized by converting values to lowercase and mapping variations such as “M”, “male”, “F”, and “female” into consistent categories.

• Similarly, the platform_usage variable was **processed** to group similar inputs into unified labels such as “Instagram”, “TikTok”, or “Both”.

Data Preparation:

• In addition, numerical values were **extracted** from the daily_social_media_hours column and converted into numeric format to improve quantitative analysis.

• Finally, unrealistic values were **filtered** from the dataset. Records with invalid academic performance values and ages outside the range of 13 to 19 years were removed to match the focus of the study.

• After preprocessing, categorical variables were **encoded** into numerical form, feature scaling was applied using StandardScaler, and SMOTE was applied to balance the classes before training the machine learning models.

MODEL SELECTION AND TRAINING

Model Choice:

In this project, two machine learning models were implemented to analyze and predict teenagers' mental health status:

- **Logistic Regression**
- **Decision Tree Classifier**

The selected models were chosen because they are widely used classification algorithms in Machine Learning and are effective for prediction tasks involving structured datasets.

Training:

The dataset was then split into training and testing sets using an 80/20 ratio. Stratified sampling was applied to maintain balanced class distribution between both sets and improve evaluation reliability.

To reduce class imbalance, the SMOTE technique was applied to generate balanced training samples before model training.

For the **Logistic Regression model**, feature scaling was applied using StandardScaler to normalize the input data, since the model is sensitive to feature magnitudes. The model was trained using class_weight = balanced to handle class imbalance and improve prediction performance.

During training, the model learned the relationship between features such as stress level, anxiety level, academic performance, and social media usage to classify teenagers into Depression and No Depression categories.

For the Decision Tree model, controlled parameters such as:

- max_depth = 3
- min_samples_split = 10
- min_samples_leaf = 5

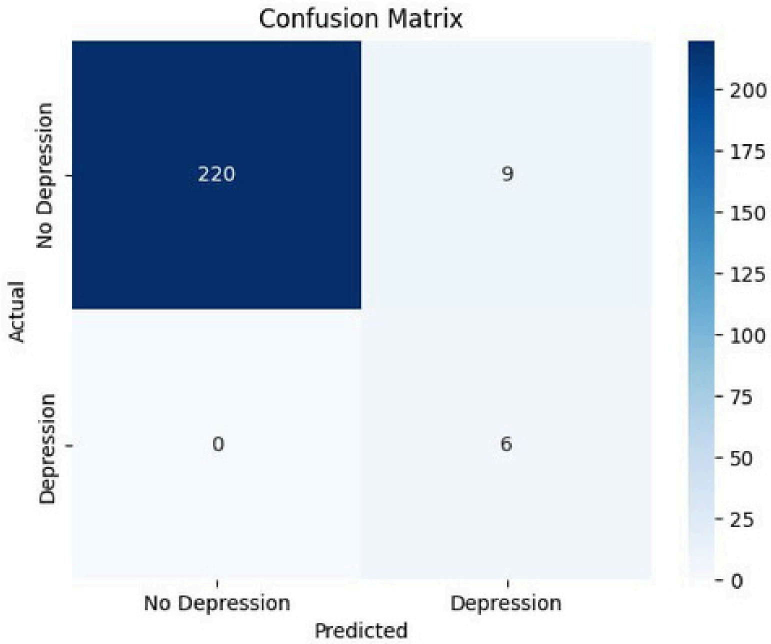
were applied to reduce overfitting and improve interpretability. The model analyzed features such as stress level, anxiety level, social interaction level, and academic performance to make classification decisions. In addition, the Decision Tree model was visualized to better understand how the model predicts depression cases.

Both models were evaluated using multiple performance metrics, including:

- Accuracy score
- Precision, Recall, and F1-score
- Confusion matrix

Finally, the results of both models were compared to determine which model provides better predictive performance and generalization performance.

MODEL EVALUATION AND TESTING



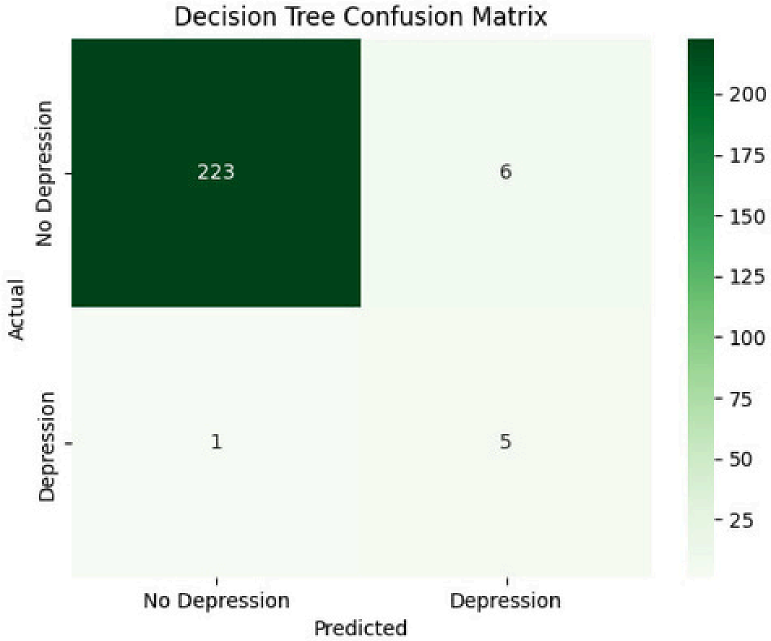
Logistic Regression Confusion Matrix Analysis:

The confusion matrix shows the performance of the Logistic Regression model during the testing phase in classifying teenagers into Depression and No Depression categories.

- **True Negatives (220)**: The model correctly predicted 220 cases as No Depression.
- **False Positives (9)**: 9 cases were incorrectly classified as Depression while they were actually No Depression.
- **False Negatives (0)**: The model did not miss any actual Depression cases.
- **True Positives (6)**: The model correctly identified 6 cases of Depression.

These results indicate that the Logistic Regression model achieved strong predictive performance with very few classification errors. The absence of false negatives shows that the model was highly effective in detecting depression cases during testing.

MODEL EVALUATION AND TESTING



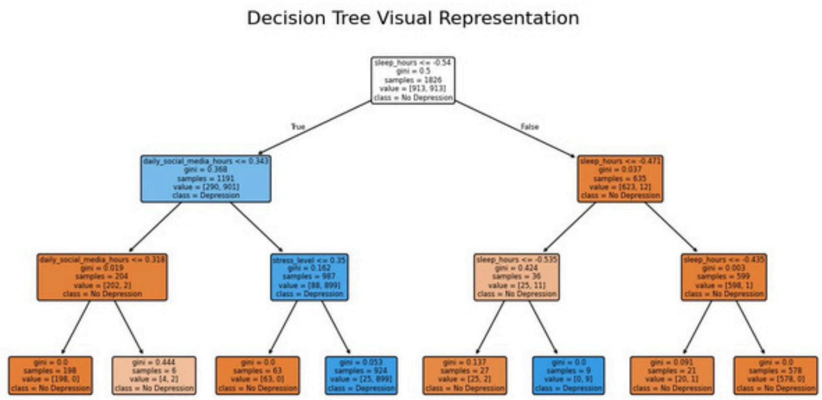
Decision Tree Confusion Matrix Analysis:

The confusion matrix shows the performance of the Decision Tree model during the testing phase in classifying teenagers into Depression and No Depression categories.

- **True Negatives (223)**: The model correctly predicted 223 cases as No Depression.
- **False Positives (6)**: 6 cases were incorrectly classified as Depression while they were actually No Depression.
- **False Negatives (1)**: The model missed 1 actual Depression case and classified it as No Depression.
- **True Positives (5)**: The model correctly identified 5 cases of Depression.

These results indicate that the Decision Tree model achieved strong predictive performance with a low number of classification errors. The model performed well in identifying both depressed and non-depressed individuals during the testing phase.

MODEL EVALUATION AND TESTING



Decision Tree Visualization Analysis:

The figure above shows the visual representation of the Decision Tree model used to predict depression among teenagers. The model classifies individuals into Depression and No Depression categories based on several important features in the dataset.

The tree begins with the feature sleep_hours, which appears as the root node, indicating that it is one of the most important factors affecting prediction results. The model then continues splitting the data using other features such as daily_social_media_hours and stress_level to improve classification accuracy.

Each node in the tree represents a decision condition, while the branches show the possible outcomes of that condition. The colored nodes represent the predicted class:

- **Orange nodes: No Depression**
- **Blue nodes: Depression**

In addition, the gini value shown in each node measures the impurity of the classification. Lower gini values indicate more accurate and pure classifications. The samples value represents the number of records in each node, while the value field shows the distribution of classes.

The visualization helps explain how the Decision Tree model makes predictions step by step and identifies which features have the strongest influence on depression prediction.
Overall, the Decision Tree provided an interpretable and effective model for understanding the relationship between teenagers' behavioral factors and depression risk.

OPTIONAL

Comparison between Decision Tree and Logistic Regression

Aspect	Decision Tree	Logistic Regression
Accuracy	97.02%	96.17%
Precision	Higher precision overall	Lower precision for Depression class
Recall	0.83 for Depression cases	1.00 for Depression cases
F1-Score	Slightly higher overall F1-score	Lower F1-score for Depression class
Interpretability	Easy to visualize and understand	Less interpretable
Feature Scaling	Not required	Required using StandardScaler
Performance	Fewer classification errors	More false positives
Best Advantage	High accuracy and visualization	Strong detection of Depression cases
Limitation	May overfit with complex trees	Sensitive to feature scaling
Overall Result	Better overall performance	Good performance with strong recall

OPTIONAL

Comparison between Decision Tree and Logistic Regression

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Logistic Regression Training Accuracy: 95.84%
Logistic Regression Test Accuracy: 96.17%

Logistic Regression Report:
               precision    recall  f1-score   support

No Depression       1.00      0.96      0.98       229
   Depression       0.40      1.00      0.57         6

     accuracy                           0.96       235
    macro avg       0.70      0.98      0.78       235
 weighted avg       0.98      0.96      0.97       235


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Decision Tree Training Accuracy: 96.80%
Decision Tree Test Accuracy: 97.02%

Decision Tree Classification Report:
              precision    recall  f1-score   support

           0       1.00      0.97      0.98       229
           1       0.45      0.83      0.59         6

    accuracy                           0.97       235
   macro avg       0.73      0.90      0.79       235
weighted avg       0.98      0.97      0.97       235
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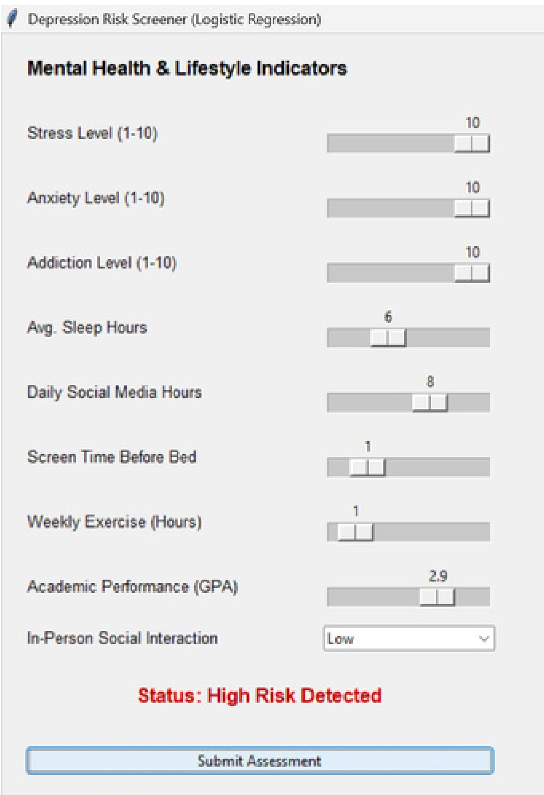
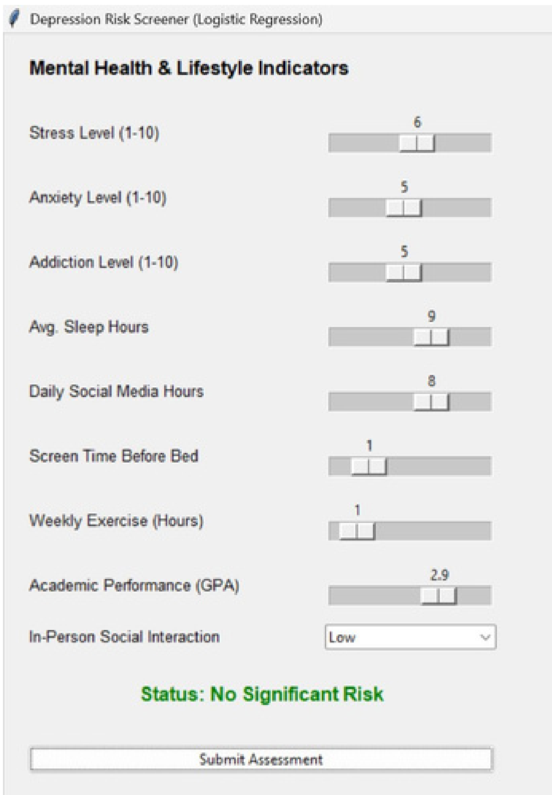
UI:





DISCUSSION

The results of this project show that both the Decision Tree and Logistic Regression models achieved strong performance in predicting depression among teenagers. The models successfully classified individuals into Depression and No Depression categories with high accuracy.

The Logistic Regression model achieved strong recall by correctly identifying all depression cases during testing, while the Decision Tree model achieved slightly higher overall accuracy and fewer classification errors. In addition, the Decision Tree visualization made the prediction process easier to understand.

Preprocessing techniques such as data cleaning, feature scaling, and SMOTE balancing improved model performance and reduced class imbalance issues. However, the dataset contained a limited number of depression cases, which may affect model generalization on larger real-world datasets.

Overall, this project demonstrates the effectiveness of machine learning techniques in supporting early detection of depression among teenagers and increasing awareness of mental health issues.

CONCLUSION

In conclusion, this project demonstrated how Machine Learning techniques can be used to predict depression among teenagers using behavioral, social, and academic factors. The implemented models achieved reliable prediction performance and showed the importance of preprocessing techniques such as data cleaning, feature scaling, and SMOTE balancing.

One challenge encountered in this project was handling class imbalance and the limited number of depression cases in the dataset. Despite this limitation, the project provided valuable experience in data preprocessing, model training, and model evaluation.

Overall, the project highlights the potential of Machine Learning in supporting early detection of depression and increasing awareness of mental health issues. Future work may include using larger datasets and testing additional machine learning algorithms to improve prediction performance.

THE END